KMT-2026-BLG-0521 / OGLE-2026-BLG-0058

Astrometric analysis — summary

Prepared 2026-09-06. Solution: `~/KMTdata/Results/v3_Final/IFfinal_260058_CTIO_.mat` (variable `IFsys`). Code at commit `47c52cf` on branch `dev1`.

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1. Data

quantity	BLG41	BLG01
exposures	19 133	19 981
after cuts	17 354	17 687
sources fitted	594	621
nights	1 985	2 009

KMT CTIO, 2016–2026, I band. Two **overlapping cut-outs** of the same star, 300 × 300 pix at 0.4″/pix (2′ × 2′). They share **no exposures**: the two fields are observed alternately, a median of 12.2 min apart, on 1 947 common nights, so they are independent measurements.

V-band frames exist for 2016–2022 (~1 340 per field) and are used only to check the colour; they are not used in the astrometry.

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2. What is in the fit

Per epoch

- full affine transformation, 6 parameters: translation, rotation, scale, shear

Per source

- position (x_0, y_0) and proper motion (μx, μy) — 4 parameters

Detrending terms

- **differential chromatic refraction (DCR)** — `[1, sin(pa)·secz, cos(pa)·secz]` plus higher orders in `secz·sin(pa)` and `secz·cos(pa)`, fitted **in 6 colour bins** (equal population). *This is included.*
- **annual term**
- **pixel-phase correction**, from the per-epoch registration shifts
- **SysRem**, 2 components on the astrometric residuals, computed **once over the whole decade** and reused by every sub-fit
- iterative reweighting from the residuals, with moving-median outlier rejection

Convergence. 15 weighted iterations before SysRem, 6 after. The last three iterations move the bright-star median by 0.003 mas (BLG41) and 0.001 (BLG01).

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3. What is not in the fit

- **parallax** — bulge sources give ≈0.12 mas, far below the noise floor
- **any microlensing or astrometric-deviation model** — the solution is model-free; no theoretical curve of any kind has been fitted or subtracted
- **absolute astrometry inside the fit** — the solution itself is *relative*: proper motions are in the registered pixel frame, 400 mas/pix is assumed rather than measured, and the orientation is not used. A Gaia tie is available afterwards (`ml.util.gaiaTie`, see section 7) and gives the orientation, the scale and absolute proper motions, but it is applied to the finished solution and does not constrain the fit
- acceleration or binary motion
- any correction for the faint-end photometric bias (§6)

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4. Steps

1. **`KMT_pipelineI`** — registration, PSF photometry, source matching → `MatchedSources`; cross-match to the OGLE catalogue for I and V, using a **per-field** pixel offset (BLG41 229.202/229.283, BLG01 227.542/229.689). 2. **`ml.util.mmsFromMatchedSources`** — magnitudes rebuilt from `FLUX_PSF` (the pipeline leaves every `MAG_` *column empty), per-epoch zero point anchored on OGLE I, then SysRem photometry; airmass and parallactic angle computed from JD and the field centre; colour from OGLE V−I; quality cuts. 3.* ***Joint fit 2016–2025*** *→ proper motions, with 2026 held out. See section 6 on why holding out 2026 alone is not enough: the event is far longer than one season. 4.* ***Full-decade fit*** *with those motions held → one SysRem correction for the run. 5.* ***Ten single-season fits*** *reusing that correction. 6.* ***Final global fit, 2016–2026, motions solved.*** This is the solution behind every plot in this report.*

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5. Calibrating stars — selection, and their colour in the plots

Selection (all conditions required):

- **14 < I_OGLE < 19**
- **no companion brighter than I = 18 within 2.0 arcsec**, i.e. **5 KMT pixels** at 0.4″/pix. The companion list is the OGLE catalogue, which is at 0.26″/pix and resolves about six times more sources than KMT detects, mapped into KMT pixels with the per-field offset above.
- a star's own OGLE counterpart is excluded from its companion test.

Surviving: **287 stars in BLG41, 253 in BLG01.**

Colour key in the RMS plots

colour	meaning
grey dots	all field stars
blue circles	**calibrating stars**, as selected above
black line	running median of *all* stars, in 0.5 mag bins
red star	the target

One point to be clear about: in this solution the per-epoch transformation is fitted from **all** sources, each weighted by its own residual scatter. The calibrating stars are a clean, well-measured subset shown for reference; they do not exclusively define the frame. (`UseRefSources`, which would restrict the frame fit to them, is available but is **off** here.)

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6. The event is not confined to one season

With the OGLE parameters t_0 = JD 2461214.2 (2026-06-22), t_E = 231.7 d and u_0 = 0.499, the magnification at each season's mean epoch is as below.

- **u** is the angular separation between lens and source at that epoch, in units of the Einstein radius: `u = sqrt(u₀² + ((t − t₀)/t_E)²)`. It is dimensionless, falls to u_0 = 0.499 at closest approach, and is large long before and after the event.
- **A** is the **flux** magnification, `A = (u² + 2) / (u·sqrt(u² + 4))`. A = 1 means unmagnified; A = 2 means twice as much light.
- **Δmag** is the same quantity in magnitudes, `Δmag = −2.5·log₁₀(A)`. The two columns are not independent: A = 2.067 is Δmag = −0.788, and −0.788 mag inverts to A = 2.066. Where the text below quotes "a magnification of about 2.1" and "0.8 mag" it is quoting the same number twice.

season	$(t - t_0)$ [d]	u	A	Δmag
2016	−3644	15.74	1.000	0.000
2019	−2564	11.08	1.000	0.000
2022	−1474	6.38	1.001	−0.001
2023	−1114	4.83	1.003	−0.003
2024	−754	3.29	1.012	−0.013
2025	−394	1.77	**1.085**	**−0.089**
2026	−44	0.53	**2.067**	**−0.788**

The event is above 1% magnification for **4.4 years**, above 5% for 2.6 years and above 10% for 2.0 years.

The model predicts 0.80 mag of brightening at the 2026 season median, whereas the measured magnitudes give 1.085 (18.724 at baseline against 17.639 in 2026). The 0.29 mag difference is the faint-end photometric bias of section 7, which is +0.39 mag at I ≈ 18.1 and +0.13 at I ≈ 17.3: a *difference* of two measured magnitudes inflates by the difference of their biases, 0.80 + 0.26 = 1.06 against 1.085 observed. It is a consistency check on the bias, not a discrepancy with the model. Calling 2026 "the event season" and 2016–2025 "pre-event" is therefore wrong: **2025 is already magnified by 9%** and 2024 by 1.3%.

This matters for the proper-motion baseline. The astrometric deviation of a microlensing event does not peak with the photometric one: for these parameters it is largest near u = √2, which falls in **2025** (2.46 mas), with 2024 at 1.91 mas and 2026 back down to 1.63 mas as the source passes closest approach. So the joint fit that supplies the proper motions, which holds out 2026 alone, still contains the two seasons carrying the **largest** astrometric signal.

Any proper motion quoted here is therefore fitted through the event, and would absorb part of a real astrometric excursion. A baseline free of it would have to stop around 2022, which costs a third of the time span and most of the leverage on the proper motion. Nothing in this report depends on that choice — no astrometric model is fitted — but a future attempt to measure the deviation must not treat 2016–2025 as an event-free baseline.

7. Caveats that matter for reading the plots

The measured magnitudes run faint at the faint end. `KMT − I_OGLE` is flat to ±0.04 down to I = 17, then rises to

+0.50 by I = 18.75. The target's own measured magnitude is 18.70 against its catalogue value of **18.13**. All plots here therefore use the **OGLE catalogue I** on the magnitude axis. PSF and aperture photometry bracket OGLE and disagree by ~0.96 mag at I = 18.5, which points to crowding: OGLE lists 3 783 sources in this cut-out where KMT detects 594. **The astrometry is not affected** — positions come from the PSF centroid, not the flux, and removing the photometric anchor entirely was measured to cost 0.012 mas.

The CSV carries two error options; choose deliberately.

column	what it is
`errX_decade`, `errY_decade`	the black running-median curve of the RMS plot read at the target's catalogue magnitude I = 18 **Constant.**
`errX_season`, `errY_season`	the target's **own** residual RMS within its observing season. Follows its brightness.

Neither is a per-epoch measurement uncertainty. The decade value is a population estimate at a fixed magnitude; the season value is the target's own scatter.

They differ by a factor of two where it matters most. The target is at I ≈ 18.1 for nine seasons and is magnified by 0.80 mag in 2026, so:

season	target mag	errX / errY, BLG41	BLG01
1 (2016)	18.73	19.28 / 18.22	19.32 / 19.51
4 (2019)	18.71	31.92 / 26.31	29.21 / 26.56
6 (2022)	18.74	33.81 / 27.17	38.55 / 31.45
9 (2025)	18.61	16.85 / 15.13	16.76 / 14.75
10 (2026)	**17.65**	**12.46 / 11.38**	**11.82 / 11.54**
decade constant	—	21.80 / 23.48	22.68 / 22.63

Using the constant would overstate the uncertainty at the peak of the event by about a factor two, and understate it in 2022 by a third.

The sign of a proper-motion component carries no physical meaning. The fitted motion of the target is +2.20 / −2.31 mas/yr in BLG41 and −0.66 / −2.90 in BLG01: the same star, the same nights, opposite signs in X. This is a gauge difference between the two frames, not anything about the source. The per-epoch transformation is free at every epoch, so it absorbs any motion common to the field, and what survives is defined only relative to the mean motion of each cut-out's own star population. Those populations differ:

quantity	BLG41	BLG01
median PM of the field's own sources	+2.128 / +0.183	−0.261 / −0.617

Measured directly on the 383 well-measured stars common to both fields, the disagreement is the same for every star:

quantity	X	Y
PM difference, BLG01 − BLG41, all common stars	−2.41 ± 0.43	−0.60 ± 0.57
the target	−2.86	−0.59

The target's offset is the population's offset. Fitting the difference as a constant plus a linear term in position gives, in addition, a relative rotation of 0.28 deg/yr and a scale drift of 4.7e-3 /yr between the two frames; removing both leaves 1.14 / 0.35 mas/yr, consistent with per-star measurement error.

So the Y values (–2.31 and –2.90) agree only because the Y gauge offset is small, and the X values disagree only because the X offset is large. **What is physical is the differential motion between stars within one field**, unless the frame is tied to an external one — which it now can be.

Tied to Gaia, the two fields agree. `ml.util.gaiaTie` carries a solution onto the Gaia frame through OGLE, giving the orientation, the plate scale and an absolute proper motion for every matched source:

quantity	BLG41	BLG01
KMT sources matched to Gaia	532 of 594 (90%)	447 of 621 (72%)
median separation	0.259 arcsec	0.358 arcsec
calibrators (Gaia PM error < 0.4 mas/yr, I < 18)	426	327
fitted rotation of the pixel axes on the sky	**+131.01 deg**	**+130.79 deg**
gauge offset removed	–0.36 / +4.41 mas/yr	–2.84 / +3.88
scatter about the Gaia relation	3.75 / 3.83 mas/yr	3.38 / 3.51

The target's proper motion:

frame	BLG41	BLG01	difference
relative, as fitted	+2.20 / –2.31	–0.66 / –2.90	2.86 / 0.59
absolute, Gaia frame	**–2.92 / –7.25**	**–2.22 / –7.24**	**0.70 / 0.01**

mas/yr. The sign disagreement disappears once both are on the same physical frame, and the two rotations, derived independently from separately reduced data, agree to 0.22 deg — which is what makes the tie credible rather than a fit to noise.

Two cautions. The Y agreement of 0.01 mas/yr is far better than either field's own precision and is therefore luck; the honest statement is that the two agree to within errors of order 1 mas/yr on the mean. And the tie fixes the **frame, not the precision**: our scatter about the Gaia relation, 3.4 to 3.8 mas/yr, exceeds Gaia's own spread of 2.9 to 3.3, so most of the per-star difference is ours and nothing in the residual analysis improves.

Plotted positions are relative to the target's own mean position, not to the field centre and not to any absolute reference. Each motion curve has the target's decade-average position subtracted, so the zero line is that average. The frame itself is set by the ensemble of 594 (BLG41) and 621 (BLG01) field stars through the per-epoch transformation, not by the geometric centre of the cut-out: differences along a curve are meaningful, the absolute level is not, and the level cannot be compared between the two fields, whose cut-outs have different origins and independently fitted frames. For scale, the target lies +0.42, +0.39 pix (+168, +155 mas) from the centre of the BLG41 cut-out and +1.55, +0.15 pix (+620, +60 mas) from the centre of BLG01; the centring absorbs both.

Only sources with an OGLE counterpart appear in the RMS plots (540 of 594 in BLG41, 478 of 621 in BLG01), which biases the plotted sample slightly bright.

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8. Where the numbers stand

quantity	BLG41	BLG01

bright-star residual (I < 17), ΔX/ΔY	6.414 / 6.879 mas	6.475 / 7.007 mas
calibrating stars, ΔX/ΔY	8.29 / 8.30	7.47 / 8.14
target, whole decade	24.62 / 21.00	25.48 / 21.78
target, 2026 season only	≈ 11.7 / 11.1	≈ 12.2 / 11.7

The target's decade figure is dominated by the seasons in which it sits near its baseline I ≈ 18.1; in 2026 the magnification of about 2.1 brightens it by 0.8 mag and its residual halves. 2025 is brighter than baseline too, by 0.09 mag. **12 mas is the target's 2026 precision, 24 mas its decade-average precision** — they are different quantities.

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9. Files in this report

file	contents
`report_RMS_afterPM.png`	residual RMS against OGLE I, per axis, per field — **after position and proper motion removed**
`report_motion___nobin.png`	source motion, unbinned; top = position with PM **retained**, bottom = residual
`report_motion___binsid.png`	the same, binned in **one sidereal month** (27.321661 d); error bars are the **RMS within the bin**
`source_motion_.csv`	`JD, X_mas, Y_mas, errX_decade, errY_decade, errX_season, errY_season, season`

Eight motion figures in total: 2 fields × 2 axes × 2 binnings.

Figures

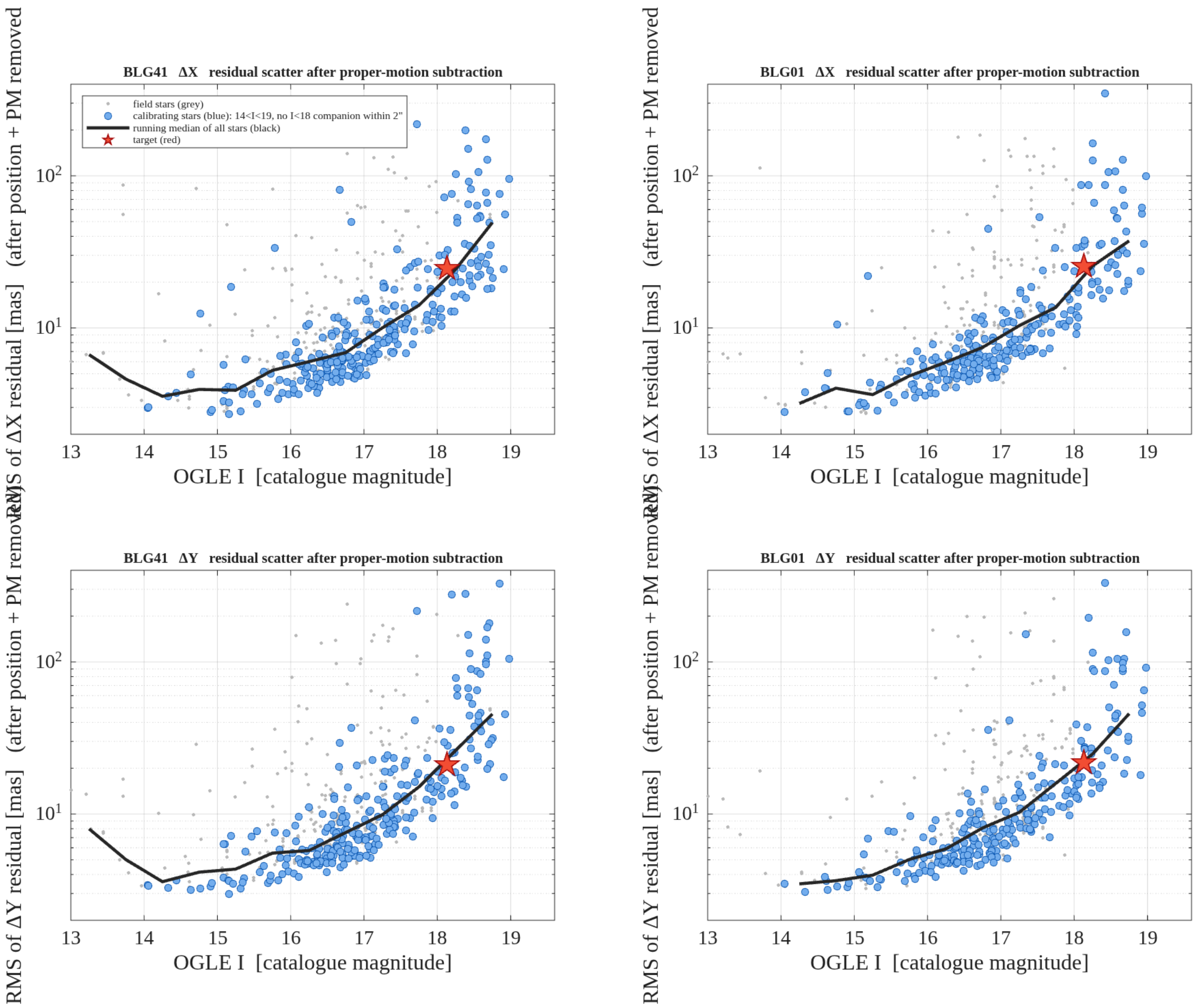

Figure 1. Residual RMS against the OGLE catalogue magnitude, per axis and per field, **after position and proper motion have been removed**. Grey: all field stars. Blue: calibrating stars (14<I<19, no I<18 companion within 2 arcsec). Black: running median of all stars, the curve the CSV decade errors are read from. Red: the target.

`report_RMS_afterPM.png`

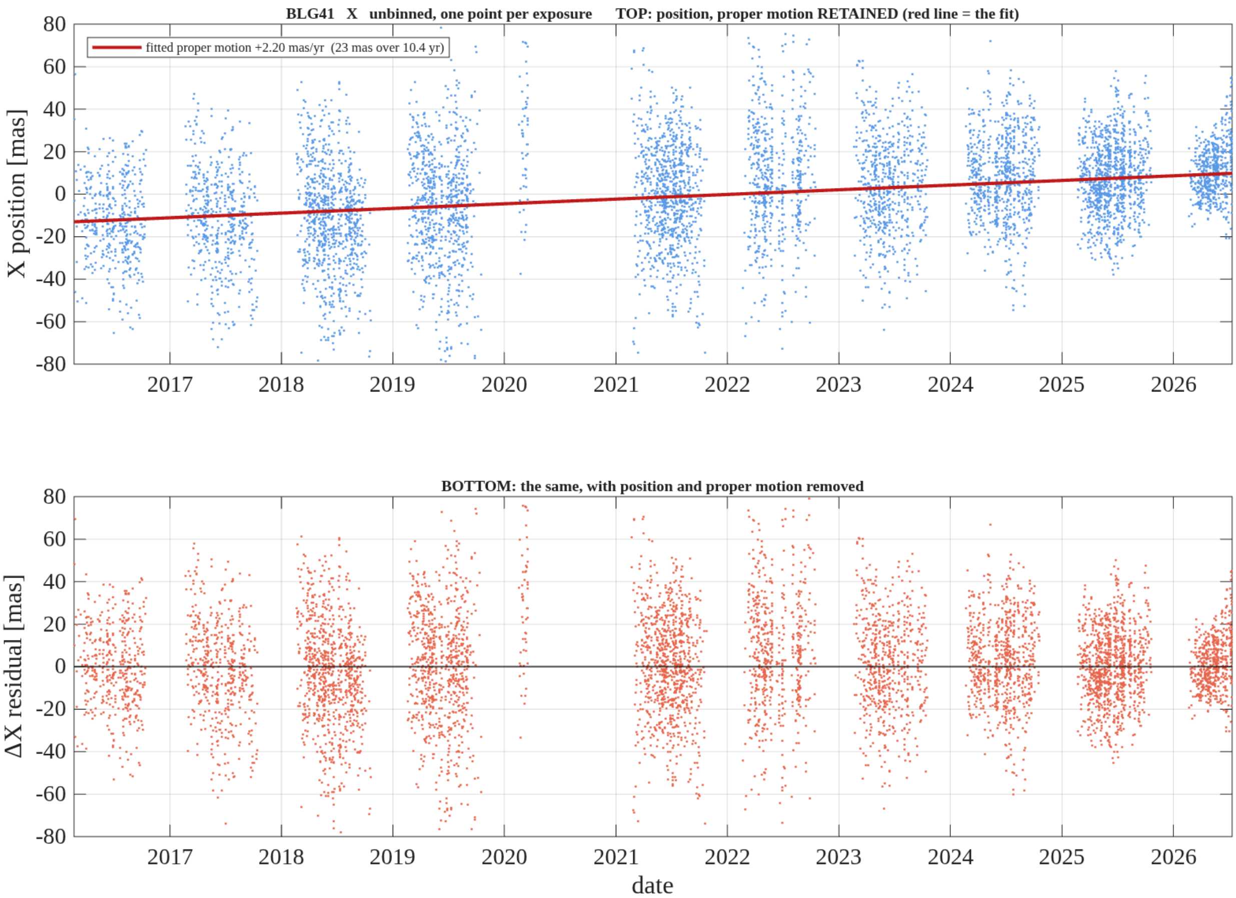

Figure 2. **BLG41, X** — unbinned, one point per exposure, no error bars. Top: the measured X position with the proper motion **retained**; the red line is the fitted proper motion. Bottom: the same with position and proper motion removed. Positions are relative to the target's own mean. The vertical range is set from the robust scatter, so a handful of extreme points fall outside it.
`report_motion_BLG41_X_nobin.png`

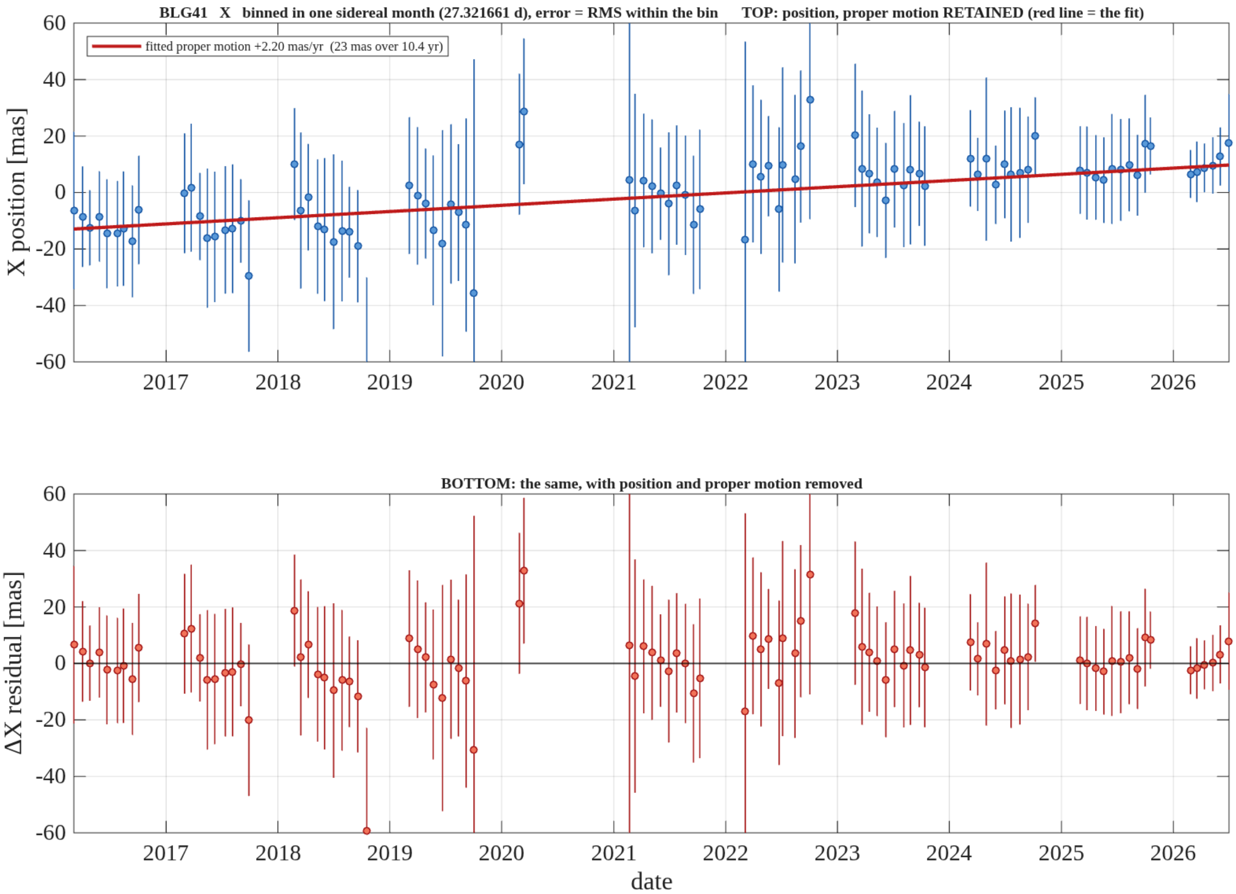

Figure 3. **BLG41, X** — binned in one sidereal month (27.321661 d); error bars are the RMS within the bin. Top: the measured X position with the proper motion **retained**; the red line is the fitted proper motion. Bottom: the same with position and proper motion removed. Positions are relative to the target’s own mean. The vertical range is set from the robust scatter, so a handful of extreme points fall outside it.

`report_motion_BLG41_X_binsid.png`

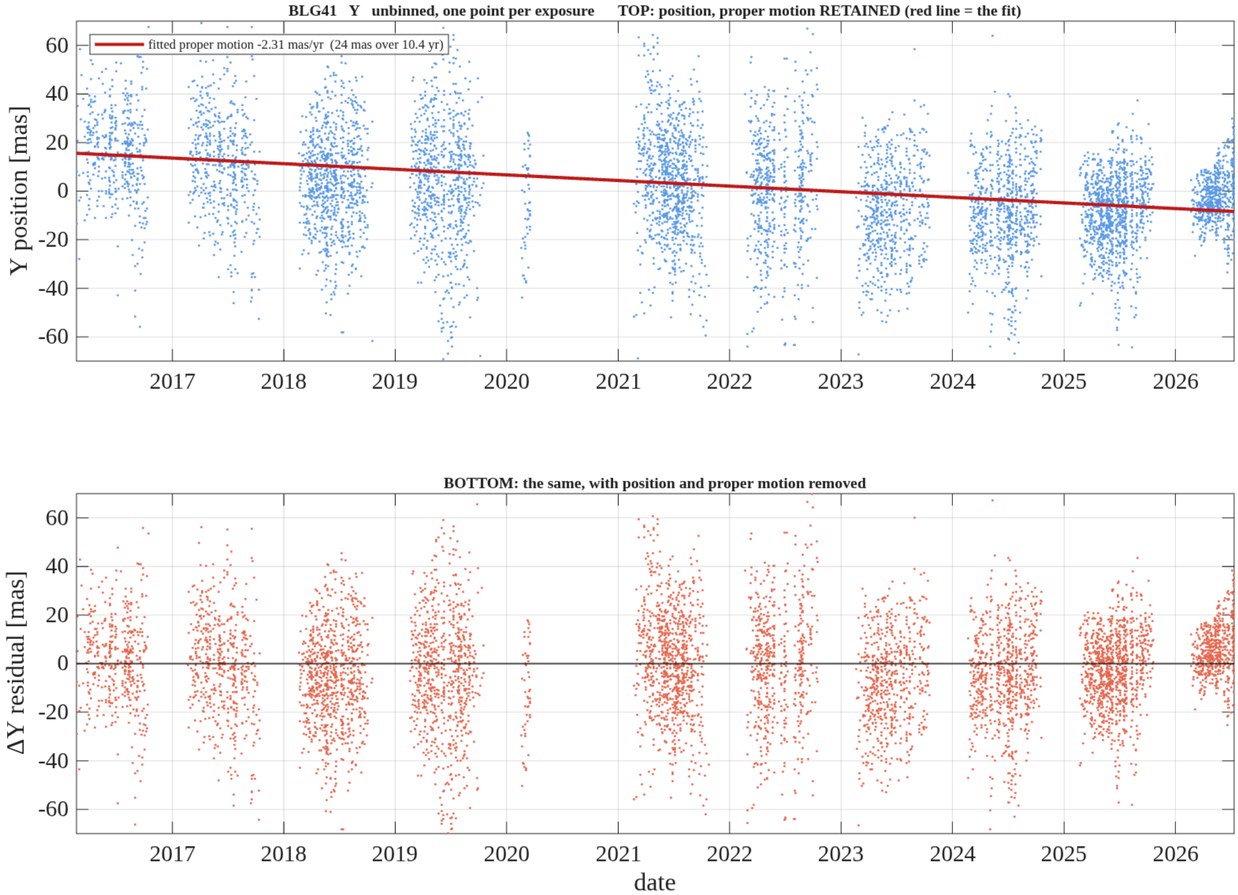

Figure 4. **BLG41, Y** — unbinned, one point per exposure, no error bars. Top: the measured Y position with the proper motion **retained**; the red line is the fitted proper motion. Bottom: the same with position and proper motion removed. Positions are relative to the target’s own mean. The vertical range is set from the robust scatter, so a handful of extreme points fall outside it.

`report_motion_BLG41_Y_nobin.png`

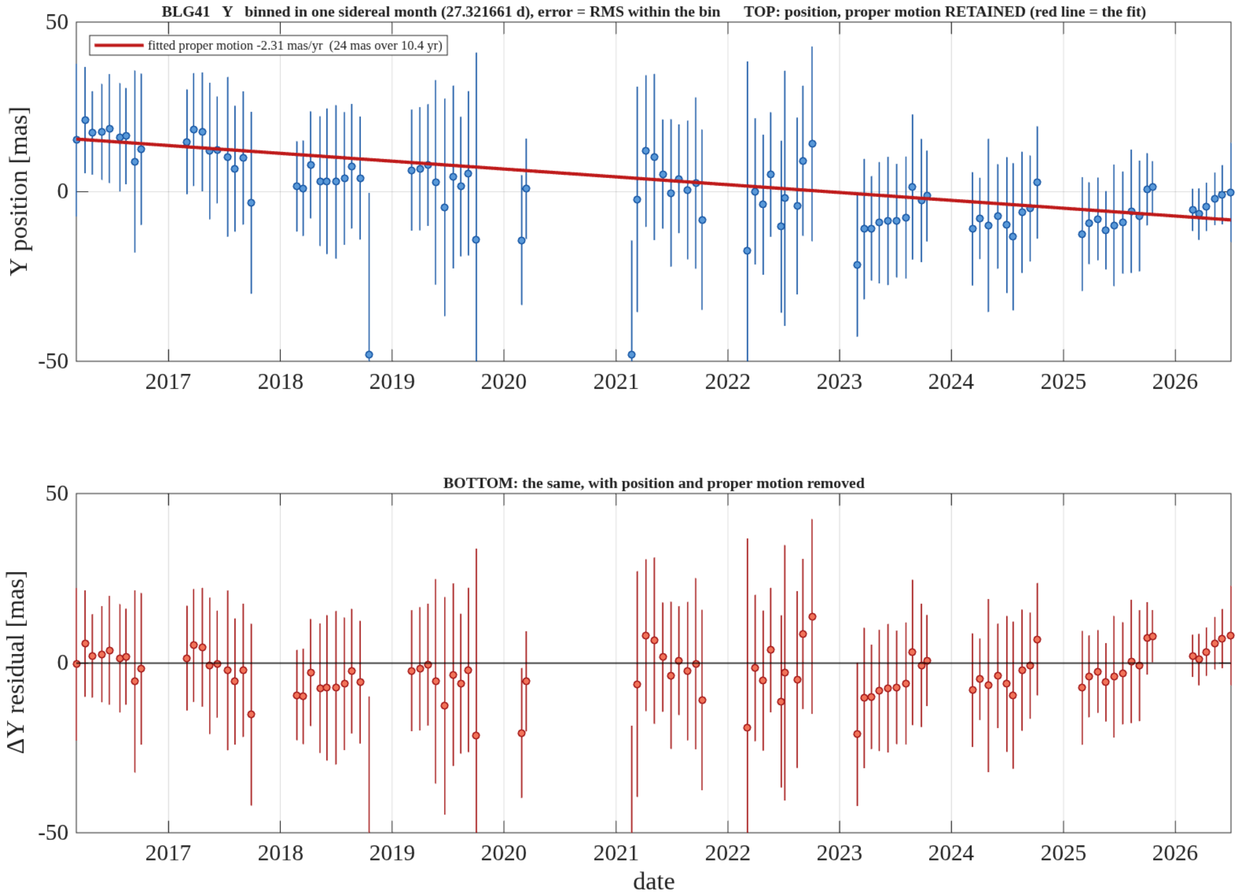

Figure 5. **BLG41, Y** — binned in one sidereal month (27.321661 d); error bars are the RMS within the bin. Top: the measured Y position with the proper motion **retained**; the red line is the fitted proper motion. Bottom: the same with position and proper motion removed. Positions are relative to the target's own mean. The vertical range is set from the robust scatter, so a handful of extreme points fall outside it.

`report_motion_BLG41_Y_binsid.png`

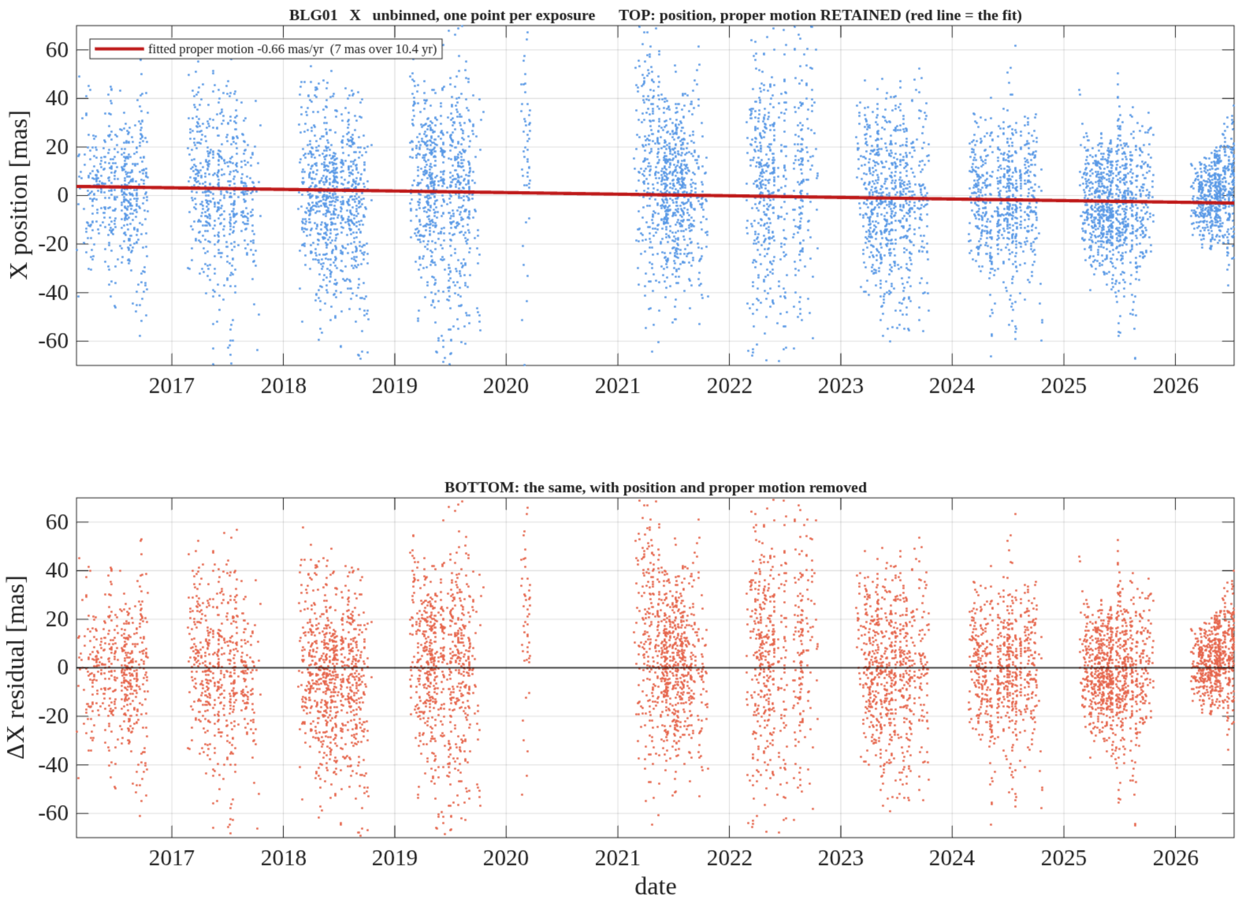

Figure 6. **BLG01, X** — unbinned, one point per exposure, no error bars. Top: the measured X position with the proper motion **retained**; the red line is the fitted proper motion. Bottom: the same with position and proper motion removed. Positions are relative to the target's own mean. The vertical range is set from the robust scatter, so a handful of extreme points fall outside it.
`report_motion_BLG01_X_nobin.png`

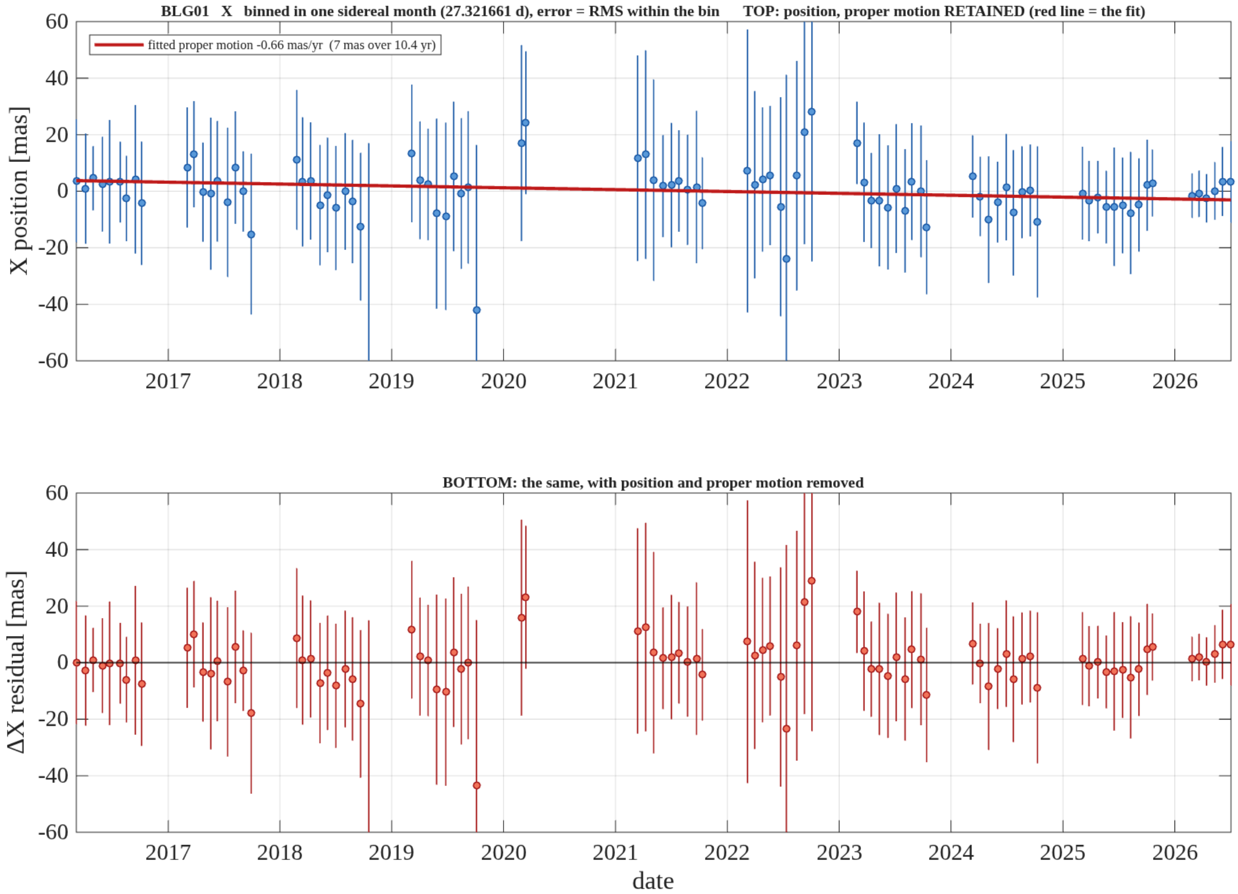

Figure 7. **BLG01, X** — binned in one sidereal month (27.321661 d); error bars are the RMS within the bin. Top: the measured X position with the proper motion **retained**; the red line is the fitted proper motion. Bottom: the same with position and proper motion removed. Positions are relative to the target's own mean. The vertical range is set from the robust scatter, so a handful of extreme points fall outside it.
`report_motion_BLG01_X_binsid.png`

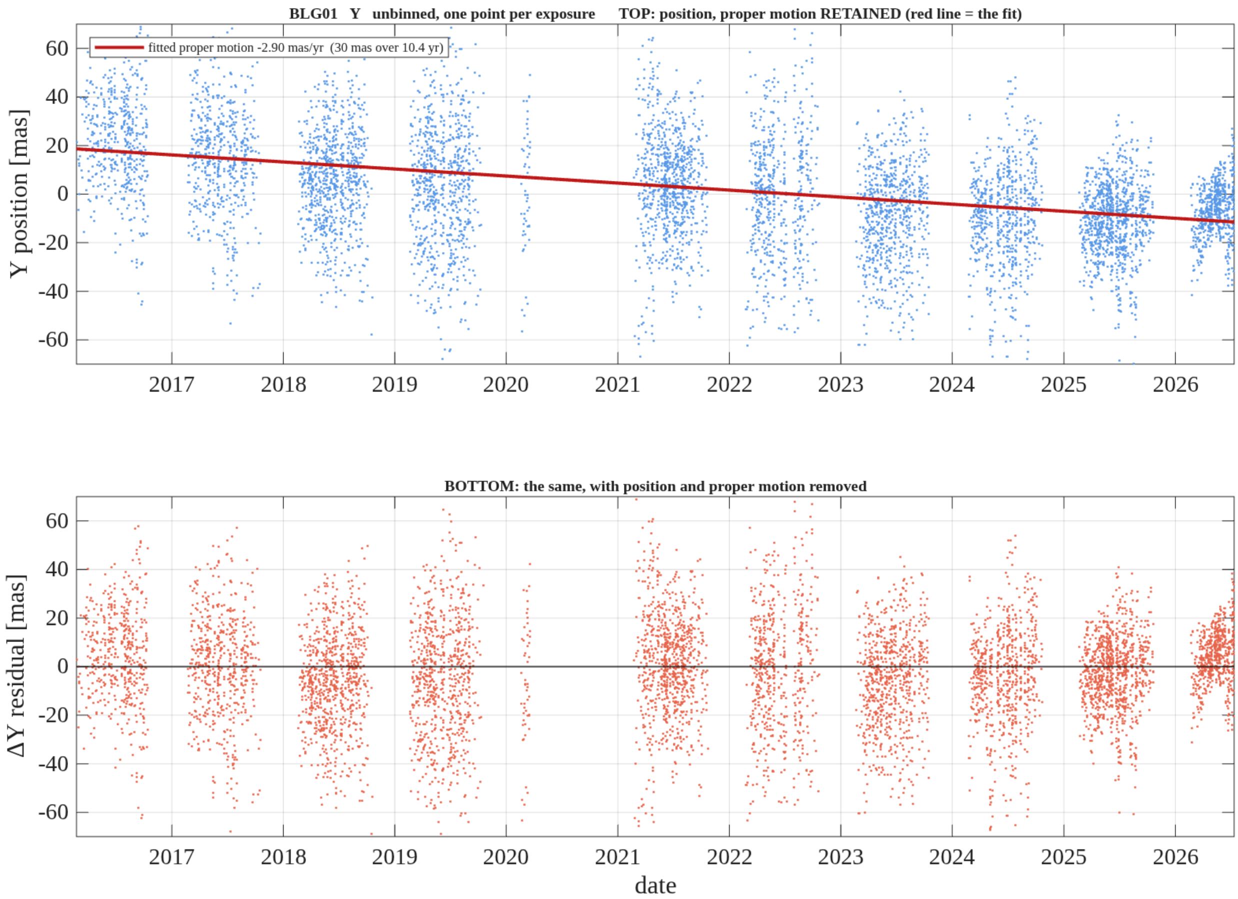

Figure 8. **BLG01, Y** — unbinned, one point per exposure, no error bars. Top: the measured Y position with the proper motion **retained**; the red line is the fitted proper motion. Bottom: the same with position and proper motion removed. Positions are relative to the target's own mean. The vertical range is set from the robust scatter, so a handful of extreme points fall outside it.
`report_motion_BLG01_Y_nobin.png`

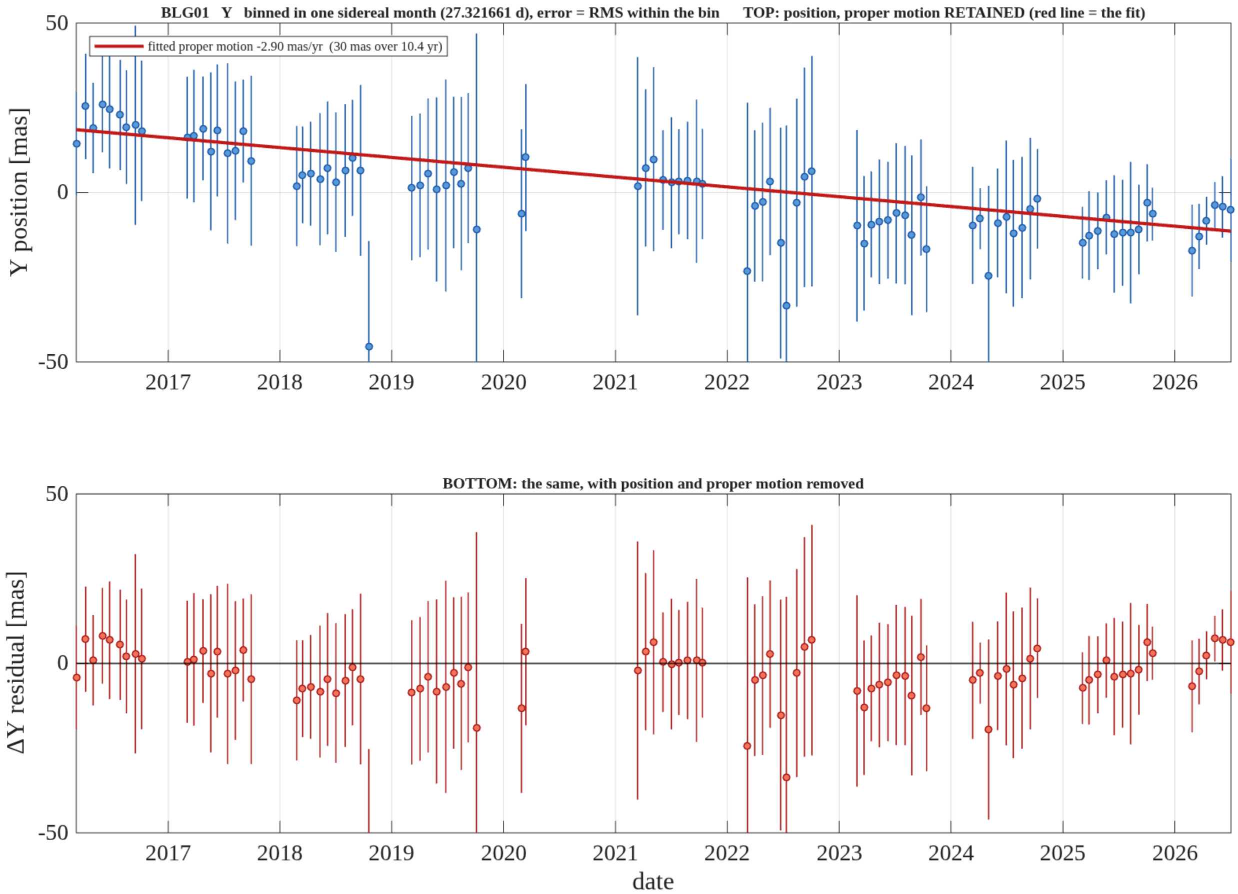

Figure 9. **BLG01, Y** — binned in one sidereal month (27.321661 d); error bars are the RMS within the bin. Top: the measured Y position with the proper motion **retained**; the red line is the fitted proper motion. Bottom: the same with position and proper motion removed. Positions are relative to the target's own mean. The vertical range is set from the robust scatter, so a handful of extreme points fall outside it.

`report_motion_BLG01_Y_binsid.png`